Number 2 — Standard form · Percentages · Percentage change

Calculator allowed. Give answers in standard form where the question is in standard form. Marks in brackets. ★ = harder question.

A · STANDARD FORM

1 Write in standard form: [4]

a $6\,800$ **b** 42 million **c** $0.000\,31$ **d** three eighths

> A number between 1 and 10, times a power of 10. **a** 6.8×10^{3} **b** 4.2×10^{7} **c** 3.1×10^{-4} (the point moves 4 places) **d** $0.375 = 3.75 \times 10^{-1}$.

2 Write as ordinary numbers: [4]

a 2.7×10^{4} **b** 9.05×10^{6} **c** 6×10^{-2} **d** 1.8×10^{-5}

> **a** $27\,000$ **b** $9\,050\,000$ **c** 0.06 **d** $0.000\,018$.

3 Work out, giving each answer in standard form (3 s.f. where needed): [6]

a $(3.2 \times 10^{3}) \times (2.5 \times 10^{4})$ **b** $\dfrac{8.4 \times 10^{7}}{2.1 \times 10^{2}}$ **c** $6.5 \times 10^{5} + 6.5 \times 10^{4}$

d $(4 \times 10^{-3})^{2}$

> **a** Numbers and powers separately: $3.2 \times 2.5 = 8$, $10^{3+4} = 10^{7} \to 8 \times 10^{7}$.
> **b** $8.4 \div 2.1 = 4$, $10^{7-2} = 10^{5} \to 4 \times 10^{5}$.
> **c** Same power first: $6.5 \times 10^{5} + 0.65 \times 10^{5} = 7.15 \times 10^{5}$.
> **d** $4^{2} \times 10^{-6} = 16 \times 10^{-6} = 1.6 \times 10^{-5}$ (16 is not between 1 and 10, so move the point).

4 Light travels at 3×10^{8} m/s. The Sun is 1.5×10^{11} m from the Earth. How long does sunlight take to reach us? Give your answer in minutes and seconds. [3]

> $\text{time} = \dfrac{\text{distance}}{\text{speed}} = \dfrac{1.5 \times 10^{11}}{3 \times 10^{8}} = 0.5 \times 10^{3} = 500$ s $= 8$ min 20 s.

B · PERCENTAGES

5 About 60% of the human body is water. How many kilograms of water are in a person of mass 65 kg? [2]

> 60% of $65 = 0.6 \times 65 = 39$ kg.

6 A laptop costs €840 before VAT. VAT is 24%. [4]

a Work out the price including VAT.

b In a sale, the price including VAT is reduced by 15%. Work out the sale price.

> **a** Multiplier for $+24\%$ is 1.24: Math input error.
>
> **b** Multiplier for -15% is 0.85: Math input error.

7 **a** A town's population grew from 2.4×10^5 in 2010 to 2.64×10^5 in 2020. Find the percentage increase. [4]

b The Earth is 1.496×10^8 km from the Sun and Mars is 2.279×10^8 km from the Sun. Write the Earth's distance as a percentage of the distance of Mars, to 3 s.f.

> **a** Percentage change $= \dfrac{\text{change}}{\text{original}} \times 100 = \dfrac{0.24 \times 10^5}{2.4 \times 10^5} \times 100 = 10\%$.
>
> **b** $\dfrac{1.496 \times 10^8}{2.279 \times 10^8} \times 100 = 65.6\%$ (3 s.f.).

8 The length of a rectangle is increased by 20% and its width is decreased by 20%. Find the percentage change in its area. [2]

> Multipliers: $1.2 \times 0.8 = 0.96$, so the area is 96% of the original — a 4% *decrease*. (Try 10×10: it becomes $12 \times 8 = 96$.)

9 A marathon record of 2 hours 20 minutes was improved by 2.5%. Find the new record time in hours, minutes and seconds. [3]

> 2 h 20 min $= 140$ min. "Improved by 2.5%" means 2.5% less time: $140 \times 0.975 = 136.5$ min $= 2$ h 16 min 30 s.

Total: 32 marks

Check: in standard form the first number is between 1 and 10 · same power of 10 before adding · percentage change is measured against the *original* · a multiplier below 1 means a decrease.